

GRIDINO, Russian Federation

WMO: 224220

Lat: 65.900N

Lon: 34.767E

Elev: 12

StdP: 101.18

Time Zone: 3.00 (E03)

Period: 82-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.8	-23.1	-27.5	0.3	-25.5	-24.8	0.4	-23.0	15.8	-8.7	13.5	-5.4	0.9	320	0.772

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	22.1	16.1	19.8	15.2	18.2	14.2	17.0	20.9	15.8	18.9	14.9	17.4	3.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.0	18.3	14.5	10.3	17.0	13.6	9.7	16.0	48.0	20.6	44.2	19.0	41.7	17.4	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
13.8	11.8	9.6	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	-29.1	19.5	4.1	2.2	-32.1	21.1	-34.4	22.4	-36.7	23.6	-39.7	25.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-12.1	-9.4	-5.2	-1.5	4.3	10.0	13.7	12.0	7.8	3.2	-4.3	-8.8
	DBStd	9.77	7.06	6.60	5.08	4.44	4.27	4.21	2.89	3.05	2.96	3.47	4.87	5.89
	HDD10.0	3601	685	543	472	346	186	53	3	14	79	212	428	582
	HDD18.3	6379	943	776	730	596	436	252	146	196	316	470	678	840
	CDD10.0	269	0	0	0	0	9	52	119	77	13	0	0	0
	CDD18.3	6	0	0	0	0	0	2	4	1	0	0	0	0
	CDH23.3	41	0	0	0	0	2	10	28	1	0	0	0	0
	CDH26.7	6	0	0	0	0	0	1	5	0	0	0	0	0
Wind	WSAvg	3.9	3.9	3.5	3.7	3.7	3.5	3.7	3.6	4.1	3.9	4.5	4.1	4.0
Precipitation	PrecAvg	459	31	25	22	22	39	42	55	58	45	45	40	36
	PrecMax	682	51	41	45	43	71	84	110	162	100	106	72	75
	PrecMin	309	8	5	5	7	11	9	5	14	8	16	2	7
	PrecStd	89	11	11	10	8	15	19	27	32	24	24	16	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	3.6	5.0	12.6	21.9	24.6	27.6	22.4	18.9	11.6	5.7	3.0
		MCWB	1.4	2.3	2.1	7.2	14.6	16.1	17.5	17.0	13.9	8.8	4.6	2.0
	2%	DB	1.0	2.0	3.6	8.9	16.4	20.8	22.7	19.7	15.6	10.2	4.2	1.7
		MCWB	0.2	0.9	1.5	5.2	10.5	14.6	16.7	15.7	12.7	8.2	3.5	0.8
	5%	DB	-0.6	1.0	2.5	7.0	13.5	18.5	20.4	18.1	13.5	8.9	2.6	0.7
		MCWB	-1.4	0.1	1.0	3.7	9.0	13.5	15.8	14.8	11.4	7.4	1.8	0.0
	10%	DB	-2.4	-0.4	1.3	5.1	10.3	16.3	18.6	16.7	12.1	7.7	1.7	-0.5
		MCWB	-3.0	-1.0	0.2	2.6	7.0	12.6	14.9	14.4	10.6	6.8	1.1	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.6	2.5	3.0	7.9	14.6	17.1	19.1	18.2	14.5	10.0	5.1	2.2
		MCDB	2.3	3.3	4.2	10.8	20.9	22.8	22.6	21.2	17.5	10.8	5.5	2.9
	2%	WB	0.2	1.2	1.9	5.5	11.4	15.3	17.6	16.3	13.0	8.4	3.5	0.9
		MCDB	1.0	1.8	3.1	8.9	15.7	19.7	21.8	18.6	15.3	9.4	4.2	1.4
	5%	WB	-1.3	0.2	1.1	4.1	8.9	13.9	16.3	15.3	11.6	7.6	2.0	0.0
		MCDB	-0.6	0.8	2.3	6.5	12.6	17.5	19.8	17.4	12.9	8.6	2.4	0.5
	10%	WB	-3.0	-1.1	0.3	2.8	7.1	12.8	15.1	14.6	10.7	6.9	1.2	-1.2
		MCDB	-2.5	-0.4	1.4	4.9	10.2	15.8	17.9	16.4	11.9	7.8	1.7	-0.5
Mean Daily Temperature Range	5% DB	MDBR	6.3	5.7	6.2	6.3	5.7	5.8	5.4	4.6	4.4	4.0	4.7	6.2
		MCDBR	6.3	4.7	6.1	8.3	11.2	10.1	9.2	7.5	7.0	5.4	4.2	4.8
		MCWBR	6.1	4.4	4.7	5.6	6.7	5.7	4.8	4.5	4.6	3.9	4.0	4.8
	5% WB	MCDBR	6.5	4.5	5.4	7.8	10.5	8.4	8.2	6.2	6.3	4.6	3.8	4.6
		MCWBR	6.4	4.2	4.4	5.8	6.7	5.2	4.9	4.1	4.6	3.7	3.8	4.7
Clear-Sky Solar Irradiance	taub	0.217	0.275	0.305	0.337	0.351	0.347	0.374	0.350	0.326	0.297	0.349	0.358	
	taud	1.951	2.077	2.139	2.253	2.420	2.488	2.450	2.489	2.528	2.349	2.584	2.604	
	Ebn at Noon	221	569	746	823	853	865	825	811	751	569	229	68	
	Edh at Noon	19	58	89	103	97	93	94	82	63	45	18	9	
All-Sky Solar Radiation	RadAvg	0.08	0.59	2.09	3.70	4.72	5.53	5.05	3.53	1.96	0.68	0.13	0.01	
	RadStd	0.01	0.11	0.21	0.29	0.35	0.50	0.56	0.34	0.20	0.10	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (9 neighbors)		+0.60	N/A	N/A	N/A	N/A	N/A	-151	-200	N/A	N/A

Nomenclature: See separate page